HW1 – Group39

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1. Results

- **Operational intensity:** Report the ideal OI in FLOP/byte, assuming a single-level memory hierarchy and an infinite number of registers.
- **Latency:** Report the median and standard deviation of the samples in milliseconds (ms), rounded to one decimal digit.

Kernel	OI (FLOP/byte)	Median latency (ms)	Std. latency (ms)
Dense scalar	42.7	592.3	12.0
Dense tiled with NEON and OpenMP	42.7	81.1	3.7
Sparse scalar	20.9	97.7	1.4
Sparse NEON	20.9	40.4	1.3
Sparse tiled	20.9	26.4	0.0
Sparse OpenMP (2 threads)	20.9	42.9	1.0
Sparse OpenMP (4 threads)	20.9	21.9	0.4
Sparse Combined (2 threads)	20.9	14.6	0.1
Sparse Combined (4 threads)	20.9	7.3	0.0

Latencies measured on a Raspberry Pi 4 (`-O0`), two benchmark runs. Median = mean of the two per-run medians; Std = standard deviation between the two per-run medians.

Operational intensity

Assumptions. OI = FLOP/bytes. Ideal model: each value is moved once (single memory level, infinite registers). 1 MAC = 2 FLOP; 1 float or index = 4 B. $M = 128$, $K = N = 512$, $nnzb = 4096$.

Dense kernels

$$\text{FLOP} = 2MKN = 2 \cdot 128 \cdot 512 \cdot 512 = 67{,}108{,}864$$

$$\text{Bytes} = 4\,(\underbrace{MK}_{A} + \underbrace{KN}_{W} + \underbrace{MN}_{C}) = 4\,(65{,}536 + 262{,}144 + 65{,}536) = 1{,}572{,}864$$

$$\text{OI} = 67{,}108{,}864/1{,}572{,}864 = 42.7 \text{ FLOP/B}$$

The tiled NEON/OpenMP version does the same FLOPs on the same data, so its OI is identical.

Sparse kernels

Each of the 128 input rows does 16 MACs per stored block:

$$\text{FLOP} = 2 \cdot M \cdot 16 \cdot nnzb = 2 \cdot 128 \cdot 16 \cdot 4096 = 16{,}777{,}216$$

$$\text{Bytes} = 4\,(\underbrace{MK}_{A} + \underbrace{16\,nnzb}_{\text{values}} + \underbrace{nnzb}_{\text{row_idx}} + \underbrace{MN}_{C})$$

$$= 262{,}144 + 262{,}144 + 16{,}384 + 262{,}144 = 802{,}816$$

$$\text{OI} = 16{,}777{,}216/802{,}816 = 20.9 \text{ FLOP/B} \quad (21.3 \text{ without index arrays})$$

NEON, tiling and OpenMP change the instructions, the order and the thread doing each MAC, but not the total FLOPs or the data moved, so all sparse kernels share OI = 20.9.

Dense vs. sparse. Sparsity cuts FLOPs by 4× but bytes only by $\approx 2\times$ (A and C are unchanged), so the OI halves.